

CURRICULUM CHANGE AND INNOVATION

CIM 8127

BY

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MODULE 1: CHANGING CONCEPTS, NATURE, PURPOSE, AND TYPES OF CURRICULUM.

Understanding Curriculum Change and Innovation

Definition of Change

- Embracing the concepts of innovation, development, renewal, and improvement in curriculum.
- A process, not an event, requires time, energy, and resources. Achieved incrementally and entails development in feelings and skills in using new programmes.
- Making the difference in some way, to give it a new position or direction. Often means alteration to its philosophy by way of its aims and objectives, reviewing the content included, revising its methods, and rethinking its evaluatory procedures.
- Leads to improvement.

Definition of Innovation

- An intentional and deliberate process to bring out desired effects and change. (Harris et al., 1995)
- Any improvement that is deliberate, measurable, durable, and unlikely to occur frequently.

Curriculum Innovation and Change

- Curriculum innovation is defined as deliberate actions to improve a learning environment by adapting a method of presenting material to students that involve human interaction, hands-on activities, and student feedback, according to the Annual Review of Applied Linguistics.

- Changes in the curriculum may involve innovation, but in general, change in terms of the curriculum involves adapting a new educational method and not necessarily a method with human interaction.

Curriculum Change

- Dictated by the changes in the economic, social, and technological aspects of society. It has magnitude and direction and takes place within a definite time frame. May occur in response to external events

Curriculum Innovation

Ideas or practices that are new and different from those that exist in the formal prescribed curriculum. It occurs when human and material resources are created, selected, organized, and used in ways where the outcomes are the higher achievement of curriculum goals and objectives. Always planned. Become meaningful and effective if they are planned and organized.

Contexts of Curriculum Change

The desire of authorities at various levels to deliberately change established practices.

- To tackle an existing problem
- Identify new problems
- Seek ways to deal with those problems

Classification of Change

- A. Hardware change -new classrooms, equipment, books, playground
- B. Software change- Affect the content of the curriculum, related to the methods of delivery of curriculum

Forms of Change

1. **Substitution change**-the replacement of one particular element of the curriculum such as curriculum guide, new instructional plan, instructional materials, or replacement of teachers and administration. In this change, one element replaces another previously in use. New textbooks, new equipment, and replacement of teachers and administrators.
2. **Alteration change**-pertains to change in curriculum structure rather than a replacement of the whole curriculum, course plan, syllabus, or instructional plan. Change in existing structures rather than a complete replacement of the whole curriculum, syllabus, or course of study.
3. **Addition change**-Introduction of a new element without necessarily changing or removing existing elements or components. Comprised of other features that may enhance the structure of the prescribed curriculum. Introduction of a new component without changing old elements or patterns. Ex: audio-visual aids, workshops, and equipment.
4. **Restructuring change**-Sharing of resources among a group of schools or institutions. Re-arrangement of the curriculum to implement. Re-arrangement of the curriculum to implement desired changes. Sharing of resources among a group of schools or institutions.

STRATEGIES FOR CURRICULUM CHANGE AND INNOVATION

1. **Participative Problem-Solving Strategy**
 - Recognizes the role of local initiative and efforts in curriculum change and innovation.
 - It focuses on the users of change and innovation, including their needs as well as how these needs are met and satisfied.

- Needs identification and analysis constitute basic approaches in employing this strategy leading to the users themselves finding the solution.

2. Planned Linkage

This strategy allows the use of what is called "intermediate agencies," such as schools, that bring together the users of the innovation.

Linkages among these users are established, and they themselves collaborate in planning how such innovation may be adopted towards more effective implementation of the curriculum in their respective schools.

The intermediate agencies, such as schools, bring together the users of the innovation in this model.

Hoyle (1993), the linkage process is based on the link between the school and the various specialized/ centralized agencies. Linkage centers may be in the form of Professional Centres, Resource Centres, ICT Centres, etc.

These agencies are the linkage point between the national agencies of curriculum development, change, and innovating schools, to provide consultancy services and offer in-service training for teachers and users.

3. Coercive Strategies

These are the strategies that are commonly used by education agencies like the Department of Education in the case of the Philippines.

Strategizing for curriculum change and innovation means the use of laws, decrees, memoranda, circulars, orders, and other similar or related documents which authorities' resort to in dealing necessarily mean "by force," but rather, it means in accordance with existing provisions as spelled out in legal documents.

Work on the bases of power and coercion by those in authority, using laws, directories, circulars, and others.

Change is accomplished through the application of power with those holding greater power enforcing compliance by those with less power.

An already-prepared curriculum, developed by government educational sectors such as the National Curriculum Development Centre in the form of a syllabus document, is given to teachers to implement.

MOE generally uses these strategies.

4. Open Input Strategies

As the term implies, these are the sets of approaches to curriculum change and innovation that make use of ideas and resources from practically all forms or kinds of sources, thus making this strategy flexible, open, and pragmatic.

CURRICULUM DEVELOPMENT AND RECENT TRENDS

Planning and Implementing Change and Innovation

Authorities and experts in curriculum development recognize the imperativeness of change and innovation that may possibly happen during curriculum implementation and as a result of curriculum evaluation.

1. The change agent

This includes the teachers, school heads, local authorities, or the Department of Education, in the case of the Philippines.

2. The innovation

This is simply the implementation of change and innovation, thus putting such change into actual use or operation.

3. The user system

This pertains to persons or groups of people at which the change or innovation effort is directed.

4. Time

Refers to a particular period of time used or spent in implementing innovation or change. Since change or innovation is a social process, the execution of change usually takes a longer period of time.

As these four factors interact with change itself, change may also happen in each of them during the process of innovation. It is also important to remember that the change agent, being the frequent user of the innovation, is involved with the change and innovation process as well as with the planning and the strategies.

As a process, change and innovation in curriculum and other related educational programs follow well-defined and specific procedures in addition to certain standards that guide in the course of implementing efforts that are intended to make the curriculum more attuned to changing needs and aspirations of the learners, there is no substitute to effective planning which considers essential requirements such as situation analysis, goal setting, formulating specific outcomes, identifying appropriate strategy or procedure in undertaking the task, provisions for building facilities and equipment needed, actual task identification and description of the social contexts within which change and innovation operate, and others like time and sequence of activities involved in the process.

MODULE 2: CURRICULUM CHANGE AND INNOVATIONS

Changes occur in the curriculum so that it can adjust to the economic, technological, social, political, and ideological needs of the society.

Types/Levels of Changes

Change can be perceived at three levels.

1. **Minor changes:** These involve re-arrangement of subject content, learning activities, re-organization of personnel, and addition of topics or methods in the curriculum project.
2. **Medium changes** involve not only the organizing of content, materials, or facilities but it involves the integration of subjects or new approaches to the existing subjects. On the other hand;
3. **The major change** involves an overhaul of the existing curriculum. It may involve complete re-organization of the conceptual design of the curriculum and change in structure, content, methods, and approaches. Materials and facilities, leading to a totally new curriculum plan or programme. For curriculum change to occur, there are certain agencies involved in the process. Let us examine some of them.

Agencies of curriculum change

Agencies of change include institutes of Education, Curriculum Development Centres, Research Institutes, Schools, Colleges, Universities, Ministries of Education (all sections and departments), Publishing firms, Examination bodies, Local Education Authorities, Teachers' Unions, and the lay public.

Importance of Change

Curriculum change is inevitable in any society. This is because: There is no perfect curriculum for all ages since society keeps on changing; hence, there is a need for curriculum to change to meet societal needs. Change links societal needs to school needs in order to bring out dynamic personalities that can address societal needs effectively. Change leads to more effective and efficient use of resources.

FACTORS THAT NECESSITATE CURRICULUM CHANGE AND INNOVATIONS

1. Social and demographic factors

The nature of the population has a direct relation to curriculum planning, development, and change. The society and the environment in which we live change, and these changes affect the schools which are part of the society. Change in the society is also reflected in the school curriculum. Social problems such as unemployment affect people's way of life. It is the curriculum that is charged with the responsibility of ensuring that social problems are alleviated.

Innovations in the curriculum may come out of a need to deal with the current population problems such as the ones you have just outlined above. Public institutions, organizations, and private agencies may be sources of change or may also support change. The type and level of support will be varied, including economic support. These institutions may also resist change if it is not in favor of the well-being of society. Changes in these institutions will generally affect the curriculum, either directly or indirectly.

2. Technological factors

In most countries, developed or developing, governments control and finance education. This means that they have a lot of control over curriculum development. A good relationship must exist between curriculum developers and the government in power. This is because the government can either promote or block any curriculum project which does not favor her policies and political ideology. The government provides support to curriculum changes and innovations. In circumstances where there are frequent government changes, a curriculum project or innovation could either take too long to take off or cease to exist. We should, however, note that curriculum projects should not contradict national policies as set out by the government.

3. Research and growth in knowledge (Epistemology)

The existing economic order in a given country such as Kenya, coupled with new technology, requires that students be introduced to new sets of skills to meet economic challenges. The state of the economy in the country will influence the curriculum. Many innovations in the school curriculum have come about due to certain economic problems. Many economic problems, such as poverty and unemployment, have forced educational systems to change their curricula in an attempt to provide solutions to the problems. A curriculum that is not in line with the official economic policies may not succeed. A new curriculum requires more trained personnel, particularly teachers, more facilities, material production, textbooks, experiments, and research to sustain the programme. This implies an increase in the national education budget. The new programme would not take off if these budgetary considerations were not put in place.

4. Political Factors

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6. National imposition

Occur when the central authority decides to adopt a new idea. Usually made known through a circular. The desire of authorities at various levels to deliberately change established practices.

- To tackle existing problems
- Identify new problems
- Seek ways to deal with those problems

7. Growth and developmental characteristics of the learners.

The emerging patterns of growth due to interrelated forces such as improved nutrition and diet, more effective immunization, etc., have led to early maturation hence the need for appropriate content to address the learners' needs. Developmental difficulties, which now include: suicide, teenage pregnancies, delinquency, drug and alcohol abuse, violence, and vandalism in schools, all have an influence on the selection of content and learning activities for the emerging learner

8. Stakeholder views

Scholarly reports of commissions, agencies, and institutes, as views and recommendations of authorities on education and related areas, may give hints on curriculum revision and change.

9. Globalization

Major role in determining the educational as well as curriculum reforms in Kenya and other countries around the globe. The revolution in information and communication technologies (ICT) made globalization possible, and knowledge became highly portable.

PLANNING AND EXECUTING CHANGE

According to Bishop 1986, there are 4 factors.

1. **The change agent** - teachers, school heads, local authorities, or the Ministry of Education
2. **The innovation** - executing the change itself; putting it into use/operation
3. **The system users**— relate to the person or group of people at which the innovation is directed.
4. **Time** – innovation is a social process that takes place over a period of time.

CURRICULUM INNOVATIONS

Innovation means the introduction of something new that deviates from the standard practice. Innovations are systematic and planned for. Innovations in education have specific and defined characteristics. Rogers and Shoemaker (1971) identified basic criteria on the basis of which the value of innovation can be judged.

Criteria for Judging the Value of Curriculum Innovation

The following are five criteria that can be used. These are: relative advantage, compatibility, Triability, Observability, and complexity. Let's examine each of them in turn.

1. **Relative advantage:** This is the degree to which an innovation is perceived as being better or introducing useful knowledge leading to acceptability and the rate of adoption. An innovation should bring improvement in the learning of students, should be economically manageable, and should have a low initial cost that poses no special risk to anybody's security.
2. **Compatibility** – refers to the degree to which an innovation is perceived as being in agreement with or supportive of the potential adopters' needs and values. Triability – This refers to the extent to which an innovation can be piloted without taking too much time, energy, or funds. The concern is whether sufficient data can be obtained from a limited trial in order to make a decision as to whether the innovation is necessary or not.
3. **Observability** - This means that the potential adopters and sponsors of innovation would like to see tangible results. The immediate consequences of the innovation should be clearly defined.
4. **Complexity** – An innovation must be simple enough to be understood and utilized. If the potential adopters perceive it as being too complicated in terms of time, money, and expertise, they may avoid it or reject it. An innovation must fit in with the goals and objectives of education, which usually reflect the needs, interests, values, and problems of society. Innovation must be appropriate, economical in terms of time, space, and resources, aligned with the philosophy of society and the school, and rooted in sound educational theory.

What, then, is involved in curriculum innovation?

Curriculum Innovation Process involves the following stages:

1. Initial disturbance: This is the pressure exerted on a system either from within or without. This disturbance constitutes the problem to be dealt with.
2. The feeling of need and the decision to act on it;
3. Diagnosis of the problem;
4. Search for a solution to the problem.
5. Application of possible solutions to solve the problem
6. Evaluation of the strategies applied to see whether they will successfully solve the problem
 1. Personnel to be employed
 2. Specification of the actual task
 3. Strategy or procedure to be used
 4. Equipment needed
 5. Conducive environment
 6. Cost and time involve
 7. Social context
 8. Sequencing of activity
 9. The rationale for undertaking the innovation
 10. Evaluation of the consequences of the innovation

Conditions for Successful Implementation of Innovations

Potential users of innovation are more likely to accept it if the conditions below are met

- The innovation must be relevant to them.
- It must be feasible in their particular organizational context.
- It must be compatible with the practices, values, and characteristics of their system.
- It must be seen as posing little or no threat to the user group's identity, integrity, and territory. The innovation must be shown to be tolerable and non-threatening.
- The innovation must yield material or non-material benefits. Gains in social status or recognition could be some of the non-material benefits.
- It must be flexible and adaptable.

MODELS OF CURRICULUM CHANGE AND INNOVATIONS

Three principal models illustrating how change takes place in the curriculum were outlined by Tanner and Tanner (1890), as cited in Curriculum Implementation (University of Zimbabwe, 1995). These are:

1. The Research, Development, and Diffusion Model

RDD_Model

In this model, an idea or practice is conceived at the central planning unit and then fed into the system. RD & D is effective where curriculum development is done on a large scale, and ideas have to reach wide geographical areas and isolated users. It is a highly organized, rational approach to innovation. Following is a logical sequence of activities using the RD & D model.

- Basic research by a central project team that develops a new curriculum and devises and designs prototype materials.
- Field trials of the prototype materials and redesign them where necessary
- Mass production of the modified prototype materials

- Mass dissemination or diffusion of innovation through courses, conferences, and workshops.
- Implementation of the innovation by the users (school, teachers, and pupils)

Summary

The process of change, which is conceptualized at the head or center, is viewed as a rational sequence of phases in which an innovation

- Invented or discovered
- Developed
- Produced
- Disseminated to the user

2. The Social Interaction Model (Periphery to Periphery Model)

SI_Model

This model operates through social interaction and emphasizes the diffusion of messages from person to person. It stresses the importance of interpersonal networks of information, opinion leadership, personal contacts, and social integration.

The strategy takes the form of convincing administrators and teachers of the usefulness of a new device or practice and enabling them to see for themselves the new practitioner using the innovation. Individuals involved are the ones to implement the innovations.

The process involves individuals rather than groups and organizations. It is unplanned and informal hence slow in development. This unplanned process can be made more systematic by structuring and coordinating the contacts between groups and individuals interested in curriculum development.

This can be done through courses, conferences, and visitations. In the social interaction model, the central agency acts merely as a coordinator or communicator of ideas rather than being the

generator of ideas. Ideas are generated at the periphery and communicated via the center to other points on the periphery.

It is based on the following-

- awareness of innovation
- interest in the innovation
- trial
- adoption for permanent use

Summary

The Social Interaction Model, stressing the importance of interpersonal networks of information, opinion, leadership, and personal contact, this model (University of Zimbabwe, 1995) describes the change as one that proceeds or diffuses through formal and informal contacts between interacting social groups.

Today's social media technology may be used as a potent venue for the exchange of views and opinions regarding efforts and initiatives towards making the curriculum a more relevant and responsive one to the changing educational requirements.

3. The Problem-Solving Model (Periphery – Centre approach)

PS_Model

This model is based on the assumption that innovation is part of a problem-solving process.

The following steps are characteristic of the problem-solving model:

- A need is identified
- Need is translated into a problem which is then diagnosed
- Diagnosis leads to a search for solutions
- Possible solutions are evaluated – the innovation which provides the best solution is then tested for its effectiveness

- Implementation of innovation.

The problem-solving method is referred to as a periphery – Centre approach to innovation. The innovations are initiated, generated, and applied by the teachers and schools on the basis of their needs. Such innovation has strong user commitment and the best chance for long-term survival.

In this model, the receiver is actively involved in finding an innovation to solve his/her own unique problem. This model is flexible enough to encompass all types of innovations, including materials, methods, and groupings of learners. Thus, the problem-solving model is local in nature, usually limited in size, and may not be of high quality compared with more centralized approaches to curriculum development.

Summary

The Problem-Solving Model is built around the user of innovation and change, this model observes the following steps:

1. Determined the problem
2. Search for Innovation
3. Evaluate the trials
4. Implement the innovation

FACTORS HINDERING THE IMPLEMENTATION OF CURRICULUM CHANGE

- **Inertia** – Unwillingness to change because people are used to the status quo
- **Uncertainty** – Feeling comfortable with the present, which is known and familiar, and reluctance to change for a future not clearly understood

- **Insecurity** – Threats of job security and status due to new knowledge explosion from the latest technology and research development
- **Limited commitment and motivation** – due to demands for more work input, new aspects to be learned, an addition to an already overloaded schedule
- **Conformity and fear** – people become less supportive if the changes are not introduced through democratic means
- **Ignorance and poor communication** – Low level of education, lack of proper information, poor training, and lack of experience at all levels can lead to poorly conceived change and poorly expressed messages, all prone to misinterpretation and misunderstanding
- **The unfavorable personality of the change agent:** The change agent can promote or hinder change depending on expertise, commitment, character status, and relationship to those whom they seek to change
- **Scarce Resources:** People may resist change if human, material, financial, and time resources are not carefully planned

CONCLUSION

When curriculum innovation or change is made in the classroom;

- It can focus on unique methods for training.
- It may encourage the creativity of trainees in the use of online tools, multimedia software applications, or hands-on lab experiments and outdoor adventures.
- Encourages educators to defy the norm and think outside of the box to reach all types of learners with individualized styles instead of focusing on lessons that are effective for just a few average students in a classroom.

MODULE 3: ASPECTS OF CURRICULUM INNOVATION IN THE KENYAN CONTEXT.

Issues and changes in curriculum are many; in recent years, we have seen a wide range of innovations emerge in education. The efforts of the innovations for the innovations are designed to improve the quality of schooling for all Kenyan children. The most recent effort has been the introduction of the 8.4.4 system of education with vocational education as its core. Other changes which have occurred in education since the early sixties through the seventies are the establishment of the Jomo Kenyatta Foundation for the publication of educational books and other related teaching materials: the establishment of the Kenya School Equipment Scheme(KSES) for the acquisition and distribution of school equipment to schools under the jurisdiction of the Ministry of Education, the scheme supplies mainly books and other instructional materials to the primary schools throughout the country.

However, many attempts to develop systems designed to meet the needs of the children; in Kenya have appeared and again disappeared throughout the history of education in the country. Although we have not yet developed die means to implement genuinely effective education programmes designed, to meet the needs of individual children, the government has increasingly a wide range of options and directions to be followed.

This lecture is devoted to a discussion of such options and changes. Some changes have been with us for a long while others are quite recent in our educational practices.

Objectives

After studying this lecture, you should be able to:

- (1) Identify curriculum changes before independence
- (2) Identify specific curriculum changes in subject areas after independence

(3) Name some of the curriculum projects that have been initiated in Kenya to alter independents

(4) Name curriculum projects initiated before independence

This study of curriculum changes in Kenya goes back to the year 1890 when Africans rule themselves. The curriculum that was offered to the youth was meant to prepare him for his responsibilities as an adult in his home, his village, and his tribe. Instructions were given by the fathers and through organized systems of elders or villagers.

The curriculum that was taught to the child included the following: Craftsmanship through an apprenticeship system, initiation of rites, religion, hunting, farming, including the raising of cattle, community responsibility, number work, dance, and music.

The elders, as part of the instructors, made sure that the youth were introduced to the legends surrounding previous exploits of their tribe, to the mysteries of their religion, and to practical aspects of running. Along this process of teaching Seaming aspects, there were varieties of formal arid, and non-formal observances in addition to the experience of daily living, which had a profound effect on the youth's place in the society in which politics, economics, and social relationships were invariably interwoven.

The period (1891-1911) of curriculum development in Kenya was heavily assisted by a few Christian missions and indirect government help given in the form of grants aid. The primary goal of missionary education was to make converts and train catechists. But it soon established as basic elements in the curriculum the following:

Practical skills, carpentry, gardening (to maintain mission stations), and literacy (reading and writing). These skills were taught specifically to children so that they could acquire skills to use and also learn how to relate themselves properly to their immediate and extended families, ancestors, their peers, and their gods.

The period (1911) marked the beginning of the establishment of the Department of Education with a Director. In 1924 there were four outstanding events that contributed largely to the process of Kenyan education. These events were as follows:

The visit of the Phelps stroke commission ate adaptation of the Education Ordinance of 1924, the appointment of the colonial office advisory committee, and the appointment of the Local Advisory Committee on African Education.

The general policy of the Educational Department as adopted in 1911 and based upon the excellent work of the great Afro-American known as Mr. Booker T. Washington in a book entitled "Working with Hands" not only remained unchanged in principles but was confirmed and strengthened first by the agriculture policy of the late Sir Robert Corydon and secondly by the principles advocated by Dr. Jones and the Phelps Stokes Commission, namely Adaptation to Environment in Education, and the distinction between the education of the masses and the education of the leaders.

By 1952 the principles governing the curriculum were based as far as possible on the Mentality, Customs, and. Institutions of the Africans. New knowledge or skill was taught in contact with the indigenous knowledge or skill. The curriculum was developed m view of the needs of the village. The life of the school provided an opportunity for the exercise of the quality of character which the colonial office wished to impart and encourage, and therefore, the curriculum was to utilize every opportunity of education arising in the life of the school.

Since independence up to the present, there has been a rapid expansion of education in Kenya. First, there was the integration of the pre-1963 African Asian, and European syllabuses into one. Then there was the New Primary Approach which was initiated in the mid-fifties by the special center. The chief developments were seen in classroom practice and in the material used for the teaching of English, Kiswahili, Mathematics, and Science by the Curriculum

Research and Development Centre formed through the amalgamation of the Mathematics and Science Centre with the special center. By 1968 Kenya Institute of Education (KIE) absorbed the CRDC and its ongoing projects. The biggest in the scale of these was the Safari English Services which were used in standards IV, V, and VI in a high proportion

Curriculum Innovations in Kenya

1. New Primary Approach (NPA)

The New Primary Approach was an innovation in the teaching of English in Primary Schools. The programme was initiated as a result of poor performance among the Asian and African children in Kenya. Plans were made to start the programme in Asian Schools first. Four years after the center was opened, the Oxford University Press launched the PEAK SERIES publications, a group of books in English designed to meet the needs of Asian children in East Africa who begin their primary education in English without prior knowledge of the language. The Ford Foundation provided printing equipment, tape recorders, and additional staff.

By 1963, the New Primary Approach had picked up very much. Teacher Training Colleges introduced the NEW SERIES English Medium. The objective of this special center was to educate through the medium of English but not to teach English. By this time (1963), the newly independent Kenya commended the programme by saying that the government had been actively attacking the problem of standards of primary education from the area of teaching methods and the curriculum. It went on to say that one of the most promising ventures in the history of education in Kenya has been the development of the New Primary Approach in primary schools. The essence of the programme is that the old concept of the child passively receiving instruction from the teacher should be replaced by a system in which the pupil develops through active and full participation in the education process.

There was a slowdown in the program by 1970 because of a lack of uniformity in the quality of the NPA program. Supervision was inadequate; classroom and teaching facilities were very poor. The government decided not to open more NPA classes.

The New Primary Approach influenced the teaching of vernacular languages, and the General Methods of KIE produced the TKK series of vernacular readers, which were very useful pamphlets on the approach to reading and a variety of other materials designed on N.P.A.-lines.

2. Science Education Programme for Africa (SEPA)

The Science Education Programme for Africa was an unusual project since its inception in 1971 when the African Primary Science Programme (APSP) handed over its responsibilities to it. The aim of SEPA was not only the production of curriculum materials, but also it attempted to build in teachers at all levels of the educational system the ability to make effective decisions about curriculum.

The project influenced the teaching of primary science in Kenya through the following objectives:

1. Promotion of effective ways of learning science by utilizing the child's environment.
2. Developing and introducing new and relevant science curricula materials into African schools.
3. (iii) Establishing institutions in Africa concerned mainly with the furtherance of the renewal of science curricula.
4. Encouraging Kenyans to write more science materials.

3. Kenyans Primary Mathematics Project (KPMP)

The Kenya Primary Mathematics project encountered a number of problems similar to those encountered by the School Mathematics Study Group (SMSG) and the School Mathematics of East Africa (SMSG)

Some of the problems that this project encountered were:

1. Negligence of making mathematics practical.
2. (ii) Isolation of mathematics from the physical and social sciences by putting too much emphasis on such topics as set theory, symbolic logic abstract, algebra, matrices, and Boolean Algebra.
3. (iii) Lack of sufficient tools for evaluation because Kenya found that the Entebbe project (SMEA), which was similar to the SMSG, failed to meet the needs of the country.
4. (iv) Lack of sufficient or proper trials.
5. (v) Teachers were not given chances to develop materials.
6. (vi) In general, this project failed to achieve its objectives.

The only major significant success that this project made was the production of plenty of mathematics materials through workshops. Also, commercial publishers entered into the business of producing many books on new mathematics.

The reading below takes us into the historical development of the KPMP and the criticisms the project received during its implementation.

This reading comes from an occasional paper written by Professor George Eshiwani. He discusses the origin of the project and gives some of the criticisms that were given to the project.

4. Kenya Primary Mathematics (KPM)

In 1964 a decision was taken to begin a long-term project to develop a new series of mathematics in Kenya Primary Schools; it was intended to use the Entebbe Mathematics series for one year in each standard, beginning with standard 1 in 1965, and then each year to replace this book with an experimental Kenya text which would lead to a definite Kenya series. Twenty-five experimental schools in Nairobi and Kisumu Districts were chosen. In-service programs for the teachers in these schools were conducted, and the Teacher Training Colleges were visited on a regular basis to acquaint tutors with the new developments. The project proceeded more or less as planned until 1968, but in 1969 shortage of staff in the mathematics section of the Kenya Institute of Education, which had been entrusted with the production of new materials, resulted in the experimental editions of the books being dropped.

In 1970 the decision was taken to extend the new series, then named Kenya Primary Mathematics, to all schools in Kenya in January 1971. The curriculum work that had to be undertaken by the Kenya Institute of Education to meet this new situation was gigantic. The KPM textbooks had to be written hurriedly and passed on to schools.

As in the case of the SMEA programme, the content in the KPM consisted of more than 60% of the topics from traditional mathematics (e.g., Highway Series). There were a few "new" topics, such as sets, bases, probability, statistics, transformation, geometry, and clock arithmetic. The major difference between the KPM texts and the traditional text was in the presentation of the subject matter.

Criticism of the New Mathematics Programme

In many countries, the new mathematics has provided a field day for cartoonists, monologists, and polemicists. Some of the criticisms have been insightful, while others have been penetrating. It is worth mentioning that several people in Kenya have written in the past about the

shortcomings of the new mathematics. Several years ago, the following criticisms of the new mathematics in Kenya were identified.

The new mathematics syllabus and the textbooks being used are far too difficult for the average child in this country, and the content is extremely demanding to both the student and the teacher. The texts are best suited to the top students, especially those who will continue in mathematics beyond high school. Little attention has been devoted to the average and below-average students. The fact that primary and secondary education is terminal for the majority of our school population dictates that the school curriculum should be tailored to this group and not to academic extroverts.

The applications of mathematics were largely ignored. Mathematics should derive from the application to the real world, or it will lose its vitality; one of the shortcomings of the new mathematics curriculum in Kenya is that it was left largely in the hands of expatriate staff who could not translate their good intentions into reality for the Kenya child.

Rigor, precision, and symbolism were overdone in both SARA and the KRM textbooks and sometimes became an end in themselves. Two effects of this were:

1. A decline in interest on the part of students whose concerns are more practical.
2. A peculiar form of notational mockery among some students, teachers, and examiners (if you cannot use the symbol, you do not know basic mathematics).

The conceptual emphasis was so great that teachers would incorporate appropriate computational skills in their instruction. The teachers did not do this, and hence the public outcry about the incompetence of the school children in performing simple computational operations.

Both the primary and secondary school syllabus was overloaded. Most teachers complained that there was so much work to be done that they had no time to try out new techniques in their teaching.

The language used in the SMRA and in the KPM textbooks is generally difficult for most pupils. This has often led to an inadequate understanding of the subject matter. Lack of curriculum diffusion between those who develop curriculum at KIC D. and teachers. The curriculum developers failed to communicate to classroom teachers what they were trying to accomplish. There were insignificant in-service training programs for teachers who were supposed to teach new mathematics. As a result, many of the mathematics festers were no better than their students. Of course, the new texts were well taught by good teachers, but in the hands of poor teachers, it was a complete disaster. There was no evaluation of the new mathematics both at the primary and secondary levels. This long-term effect of changing from one type of curriculum to another was therefore not evaluated.

The Gachathi Report of 1976 spelled out in detail the objective of education. This commission was assigned the task of investigating the quality of education in Kenya. Primary education has been accused by members of the community it lacked quality, content, and relevance, and therefore, it was not catering sufficiently to the majority of our children for whom primary education is terminal.

In an effort to fulfill this responsibility, the government decided to introduce far-reaching changes in Kenya's Primary Education Programme (PEP), which took into called primary and relevant. The development of PEP started in 1997, and in progress was the piloting of materials for standards one to five. Piloting of other classes followed this until they were all covered. The development of PEP was based on the premise that:

- (i) There is a need to improve the quality, content, and relevance of primary education so that it centers more effectively on the majority of children for whom primary education is terminal.
- (ii) Primary education should be made available to all primary school-age children.
- (iii) Primary education should be broadly based and lead to the development of competencies in a variety of practical skills.
- (iv) Primary education should concentrate on the needs of the majority who terminate their education at the primary school level while bearing in mind the needs of those who will continue to the secondary and tertiary cycles.

The programme consisted of three phases:

- (i) Designing a primary and primary teacher education curriculum programme and production of curriculum materials.
- (ii) Implementation of the primary and primary teacher education programme into all the Primary schools and primary teacher's colleges.
- (iii) Summative evaluation of the programme.

Mackay's Report of 1981, which was the presidential working party on the second University, recommended, among other things, major changes to the 8-4-4 system of education.

The essential elements of the new system are in the area of the structure, which has 8 years of primary, 4 years of secondary, and 4 years minimum university education. In the area of curriculum, content is oriented towards technical education and not examination centered.

The preparation of the curriculum for the 8 years primary cycle was patterned on similar lines to the primary education project popularly known as PEP, which was by then being developed by the Kenya Institute of Education.

The most significant aspects of PEP with the 8 years of primary education are:

- (i) Focus on the entire primary education.
- (ii) Renationalization of primary education
- (iii) Reorganization of subjects into broad curriculum areas with common objectives
- (iv) Introduction of new subjects to meet identified needs not presently adequately catered for.

The curriculum is organized on two broad cycles:

- Lower primary cycle (Std.1-3)
- Upper primary (Std.4-8)

Lower primary (Std. 1-3)

The main emphasis at this level is the development of numeracy and literacy language.

- (i) Mother tongue (including Kiswahili, where it is used as a mother tongue).
Mother tongue is also used as the medium of instruction.
- (ii) English

Mathematics

Science (integrating agriculture, home science, and general science).

Social Studies (integrating geography, history, civics, and social education).

Religious Education

- (i) Christian Religious Education
- (ii) Islamic Religious Education

Creative Arts:

- (i) Art and Craft
- (i) Music
- (ii) Physical education

Upper Primary (Std.4-8)

The main emphasis at this level is the development of practical skills to prepare children for the world of work. The subjects to be followed are:

Languages:

- (I) Kiswahili
- (II) English - also used as the medium of instruction

Mathematics

Science

Home Science

Social Studies (incorporating Geography, History, Civics, and social education).

Religious Education

- (I) Christian Religious Education.
- (II) Islamic Religious Education.

Practical skills education

- (I) Agriculture.
- (II) Art Education.
- (III) Craft Education.
- (IV) Business Education.

Music

Physical Education

In an attempt to relate primary school education to the development of life-oriented skills, a new cluster of subjects known as practical skills was introduced in primary education; this was

adapted after extensive consultation and discussion in preference to occupational skills education that had been used earlier. The latter is considered to be presumptuous and gives the impression that at the end of primary education, the pupils will have prepared for a job or occupation. The most that can be expected to be achieved within the primary cycle is the building of a solid education foundation with some practical skills which the pupils can build after leaving school.

a). Standard 8 Curriculum

According to the primary education (PEP) project plan for Std. 8, piloted in 1986, it was to be available to all schools until 1989 at the earliest. Therefore, to facilitate the implementation of the 8.4.4 program in 1985, an interim curriculum program for Std. 8 was replaced by PEP materials until it was completely phased out.

The interim curriculum was based on existing curriculum areas, that is, the subjects as they appeared on the current primary timetable and not as proposed by PEP. However, they were not unduly penalized as they waited for the completion of the development of PEP materials. Care was taken to ensure that the transition from std. 7 to 8 in the interim period was smooth.

A new examination was developed and used in 1985. The details of the examination were worked out alongside the development of the interim curriculum. A new examination was also developed through PEP.

b). Secondary Education Cycle.

The presentation of the secondary education cycle was made by the Director of Higher Education, who made the following:

That secondary education should take off from the cut off from the point of primary education and hence down to standard VIII.

- (i) That current from own work should not be pushed down to standard VIII.
- (ii) Stds. VII and VIII will need to be diversified.
- (i) The development of an elaborate secondary education curriculum is to take into account the basic preparation offered in the primary education cycle. Hence it has to await the completion of the development of the 8-year primary education curriculum.
- (ii) The secondary education curriculum is to be based on the terminal level of the primary education cycle.
- (iii) There was a need for continued close cooperation and liaison between the Ministries of Basic and Higher Education in the implementation of the 8-4-4 education system.

That there was a need for an agreement with the University on the terminal level of secondary education, particularly on what aspects of the current KACE academic work should be retained in secondary education and what aspects should move to the University.

There was a need to examine the various issues relating to the secondary terminal examination and certification, as well as the nature, level, and duration of the various post-secondary training programs.

Four-year University Education Cycle

The vice-chancellor of the University of Nairobi made the following presentation on the University Education Cycle. He made the following comments:

He reported that the senate, in its discussion last year on the 8-4-4 education, had raised several points regarding,

- (i) The level at which secondary school leavers will be selected for university entry.
- (ii) What would be covered in the extra university year?

He noted that university education in East Africa is currently based on 'A' level academic preparation and that recommended change to 4-year secondary education is very fundamental.

The working party on the Second University did not go into details about the changes in University Education. It recommended that it should be more practical in its approach. Graduates were to be practical-oriented and aware of general development issues and strategies. It was also recommended that the B.Ed. The programme at Kenyatta University should be reviewed to make it richer in content.

He noted that while University Education must continue at the present level of competence, there were mundane implications of lengthening the duration of University Education.

Summary

In this lecture, we have been able to discuss curriculum changes that have taken place in Kenya's educational development. Specifically, we have stated that changes in curriculum in Kenya go way back to the pre-colonial and colonial eras. In 1963, the New Primary Approach was very popular as an innovation in schools. NPA did not last long in the system due to a number of factors.

Science Education programme for Africa, the Kenya Primary mathematics project, and other similar projects were introduced after independence.

An important change that has occurred in the history of our educational development in Kenya has been the introduction of the 8:4:4 cycle system of education.

Activity

- (i) Discuss some; curriculum changes in Kenya between 1960 and 1966
- (ii) Read more on NP A and give your views on why the die project failed.

- (iii) Examine the criticisms against Kenya's primary mathematics. Why was the programme criticized so strongly?
- (iv) Which are the areas in the 8:4:4 syllabus that you consider to be new, and which ones are not? Give reasons for your reactions.

Forces Affecting Curriculum Innovation in Kenya

People who develop curricula are faced with a lot of issues to wrestle with. Some of these affect curriculum changes. In this lecture, we shall refer to these issues as forces that affected curriculum changes in Kenya.

Some specific forces have been selected for our discussion in this lecture. There are many others that could not be discussed in the small space for this lecture. We shall particularly discuss the drive for power in a force, the appeal for the shillings, growth in knowledge, and people's needs in schools.

Objectives

After studying this lecture, you should be able to:

- 1) Describe some factors which affect curriculum changes.
- 2) Explain how growth in knowledge has influenced curriculum decisions.
- 3) Examine various needs of the society, which must be considered in curriculum planning

Curriculum development is a difficult and complex task. There are many problems and no ready solutions. In many curriculum cases, one would find that our schools appear to stress the teaching of subject matter (knowledge) and forget to remember that the child's needs are paramountly. There is too much class instruction going on in our schools and too little education on the hands. The question of how far our curriculum is in tune with our social change, needs, and future aspirations has been asked again and again. No solution has been given to this question. Those involved in the construction of the school curriculum tend to

adopt foreign ideas and use them in teaching. Some of the ideas become completely impracticable and are abandoned before they mature.

There is a need for changes that would occur gradually and not abruptly. In most cases, the syllabus is designed in such a way that knowledge is brought to the child who goes seeking knowledge. The teacher should initiate the child and arouse his curiosity which will lead the child to seek knowledge. This process can be termed self-learning, which is true learning. President Moi has repeatedly pointed out that discipline should be studied more than the content of the subject. This is to say that in studying agriculture, children should study the subject from the agriculturist approach and so on. The exploration and inquiry approaches should be given priority. While studying a discipline, it should be studied in the manner in which it affects our society.

Our classroom instruction does not provide sufficient freedom to the child. A flexible classroom is needed where children go on discovering the world around them as they seek that knowledge that is unknown to them.

From the beginning of Kenya's independence in 1963, hostile influences have continued to plague our school systems. Groups and individuals of varied viewpoints have since then affected our schools. Today those of us in curriculum work have recognized some hostile agents, and thus some force themselves to affect the curriculum intimately and consistently. These special and permanent forces, with their temporary agents, tend to cause curriculum change, though sometimes they hold it back. Because the curriculum is where people are, the special, permanent forces bringing about the other affecting curriculum changes are dearly human. Each force, in its quality of humanness, holds potential for good and potential for the end. Each lies deep inhuman motivation.

Four forces affecting curriculum in Kenya have become especially prominent. These are:

1. The drive for power.
2. The appeal of the shilling.
3. The growth in knowledge with corresponding efforts at evaluation and acquisition of knowledge.
4. The needs and concerns of people in schools within surrounding social and cultural factors.

1) The Drive to Power

During the sixties, the people's drive for power over the curriculum revealed itself in the urge of people to speak loudly, to alert other citizens to an alleged problem, and to become nationally prominent. Sometimes the drive had a helpful end; often, it seemed only a quest for power for people's sake. During the early sixties, the attack on the curriculum was justified because the colonial curriculum was not designed to meet the needs of the Kenyan people. So improvement was needed in providing education geared to meet the need of the learners and for national development.

There has been in the past militancy by teachers' organizations which have learned that when one begins to talk about teachers' welfare, he must soon discuss the organization of schools and children's curricula, both of which matters have previously been in the presence of the Ministry of Education.

The militant behavior of youth, beginning at University and moving to secondary schools. There has also been a push by scholars in the subject fields or political positions who, often, at the expense of professional educators and especially curriculum leaders, criticize what primary and secondary schools are teaching.

While formal arrangements for decision-making about the curriculum have not materially changed, the people who have initiated and sanctioned curriculum ideas have often been those who do not understand the concept of curriculum change.

2) The Appeal for the Shilling

A second fundamental force that has affected curriculum change in Kenya has been the strong appeal that money has for curriculum makers. People are always in need of funds to do what they have wanted to do for children; curriculum personnel has found a bonanza in grants-in-aid, which have frequently proved to be mixed blessings. The Ministry of Education has become a seeker of special grants or loans to improve the curriculum. When this money has been acquired, the government, through the Ministry of Education, still has something to say about ways in which funds are to be used for supporting and expanding the curriculum.

On the other hand, foreign donors in recent years have frequently earmarked, designated, or categorically controlled the precise nature of curriculum reform they want. This has been viewed by the country receiving the aid as stifling creativity and holding back development in the third world, among which Kenya is one, as well as causing excessive dependence on the developed countries.

While grants or loans have emphasized, for example, particularized the teaching of new mathematics, material developers have demanded financial profits through new educational ideas and an increase in the child population. Because of the appeal of the shilling, the producers of educational materials have flooded the market with these materials, which are conditioning increasingly what children learn. Thus, one can say that both the curriculum package sealers from foreign countries and the sales promotion schemes of businessmen here in Kenya are having unprecedented impacts on curriculum decision-making.

3) Growth of Knowledge

A third force persistently affecting curriculum change is growth in knowledge, which in the past occurred slowly and quite steadily but now shows a marked erratic burst of speed. The teacher is no longer able to recover the book. Instead, many books now cover the teacher. Nowadays, knowledge filters in all fields, so Herbert Spencer's question of "What knowledge is of most worth?" becomes more and more pertinent. Against a backdrop of educational objectives, curriculum planners are forced to seek new answers to Spencer's questions.

4) Needs of the People in School

A fourth force affecting curriculum change is the need and concerns of pupils, teachers, parents and administration for children.

The real needs and concerns of people have part of their foundation in society and community. Therefore, parents and other community members should be expected to contribute to in-school education. This is not the case with our Kenyan community where most parents feel that the school is an independent institution away from the society in which they live. The present economic order in Kenya assisted by a new technology requires that

Pupils be introduced to new sets of skills to deal with it. The experience in the urban and the rural call for school curriculum to face live social problems. Furthermore, the current crisis in unemployment and values among our youth makes us seek better ways of educating in the effective domain.

A major concern of teaching and administrators is for pre-service and in-service development of teaching skills which will help teachers do their best in classrooms. Because of not providing better training facilities to our teachers, it has been difficult for these curriculum change.

Summary

This lecture has explored various issues and changes that have taken place in Kenya before and after Idler's independence. Among three issues and changes that took place before independence was died Phelps Stokes report, which recommended separate education for Africans, Asians, and white children allowed to proceed to secondary schools.

The post-independence curriculum changes are discussed starting from the Ominde report (1964), die Gachathi report (1976), and the Mackey report of 1981. All these education commissions and committees did recommend that Kenyan education should be made more practical for the Kenyan child whose education is likely to end at the primary school level.

Curriculum change as they have wished. In part, they have been able to show interest in ideas that can now be seen in the newly introduced 8-4-4 system of education. Experience has shown that to organize human knowledge for teaching; there is a need for academic scholars to team up with curriculum specialists, behavioral scientists, and specialists, behavioral scientists and specialists in research and evaluation of curriculum aspects.

With the increasing growth of knowledge, definite attitudes towards its growth have emerged. While one of those attitudes has been concerned with how to sort out elements of knowledge and place them with the curriculum, another attitude has been fear that even the former elements are not being understood and learned.

Some curriculum development tips

Role of curriculum agents, steps in the curriculum process, translation of theory into practice, recognizing and rewarding academic excellence, affecting methods to enhance learning, the need for the foundation of curriculum, classification of curriculum objectives, behaviorism in the classroom situations, teaching critical thinking, deciding and applying teaching methods, understanding curriculum designs and sharing curriculum activities.